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Branding is what people say about you when you are not in the room.



-Jeff Bezos



Second, assuming there are markets where the standards compliance is lax and the products can be sold, risk of safety issues and the litigation due to that can far outweigh the compliance to standards. Complying to standards must start from design stage and not after the design is complete. Let us see which activities get impacted due to standards' non-compliance in the Table.

Some examples

Just to show the importance, two instances of compliance to standards not being met are briefly described below along with the mess they created.

First, we were designing an advanced antenna that could adjust its lobe width by operating a motor mechanism. We completed the testing and certification. However, when this antenna was installed in a European nation known for low temperature, we noticed two things. One, the motor froze, and second, the processor failed to start. In both cases what we found later was that the components we had used for testing and those used for manufacturing (even though the datasheet said they were equivalent) were different, leading to the failure. While we could sort the motor issue, we found that most crystals don't start at -15°C, so we ended up preheating the component used in the design to ensure there were no issues.

Second was a networking product, which was packed in small wooden crates to be shipped to Europe. When the contents landed in European airport the authorities sent the material back due to pests and decay in the wood used. We had to import the crating wood for packaging! This taught us a lesson that even a small miss can cost lots of money and time.

What need to be factored

There are three critical things that need to be factored in when de-

signing products for compliance to standards:

1. Know the country the product is supposed to be sold in and get to know the compliance laws and standards well. If the product is being designed in one country and exported to another, the best way to manage is to use global certification agencies like TUV to get the certification as they have the experience and knowledge to address global standards.

2. Spend time in selecting components and their equivalents. Above all, even if the product has custom parts made by suppliers, the responsibility of those parts meeting the standards is always yours. Never leave it to the suppliers as they invariably don't appreciate the need for complying to standards. (An entire lot of advanced remote access server we shipped to the US was sent back as the welded portions rusted while they were in US airport for a week.)

3. It is advisable to have a full-time engineer to address the issues if the products are being targeted for global markets. This helps in keeping the compliances issues to the minimum.

Points to keep in mind

Complying to standards calls for a structured approach. In every phase of the product development, impact of standard compliance has to be assessed and actions have to be taken. This is very important. If a list of the items to be checked is prepared, it will run into a book. For a reliable and very useful book on the subject, check <https://emcfastpass.com/certificatebook>. It is a good book for MSMEs and startups to understand global standard compliance and how to deal with it. But remember, it addresses all the electrical standards. Designers have to understand the standards for other certifications as well, especially the environmental ones. **EFY**